# Lecture - 8

# Chapter 4

# FILTERING IN THE FREQUENCY DOMAIN

# Fourier Series

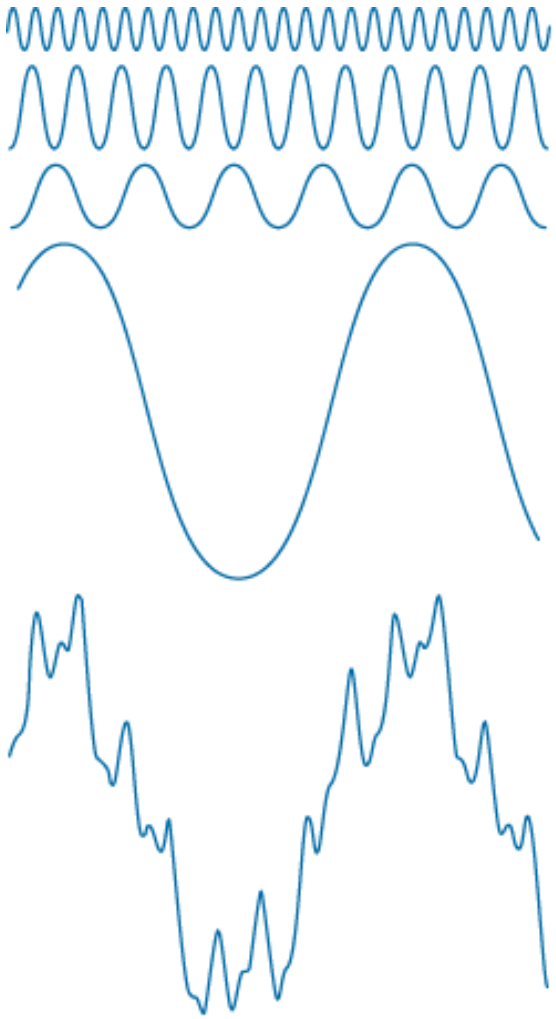
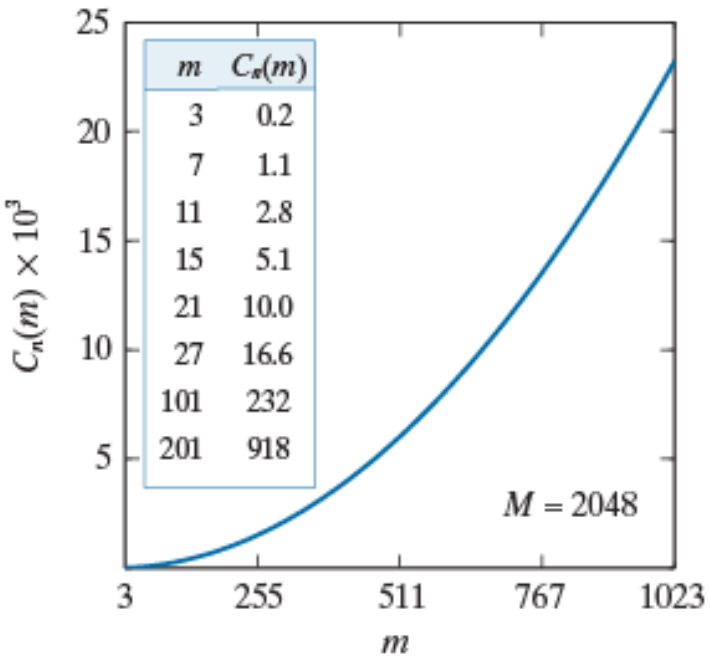
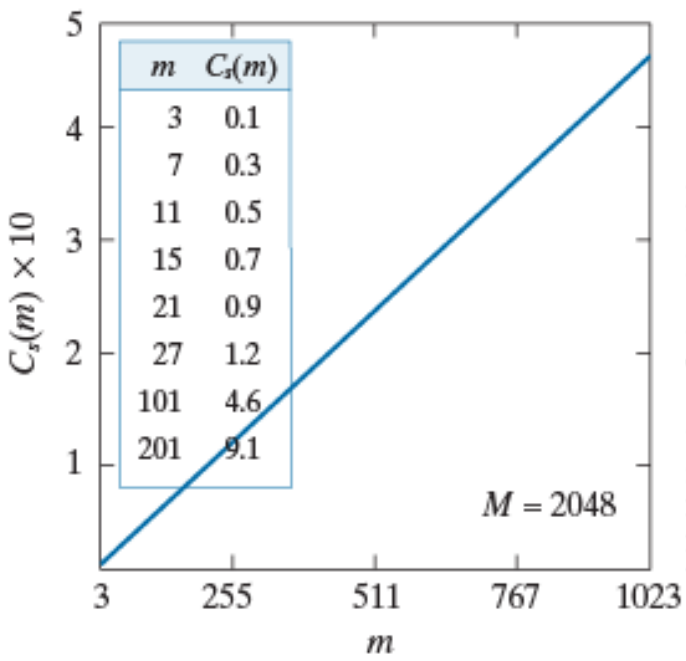


**FIGURE 4.1** The function at the bottom is the sum of the four functions above it. Fourier's idea in 1807 that periodic functions could be represented as a weighted sum of sines and cosines was met with skepticism.

# FFT Computational Advantage over Spatial Filters




| $m$ | $C_n(m)$ |
|---|---|
| 3 | 0.2 |
| 7 | 1.1 |
| 11 | 2.8 |
| 15 | 5.1 |
| 21 | 10.0 |
| 27 | 16.6 |
| 101 | 232 |
| 201 | 918 |




| $m$ | $C_s(m)$ |
|---|---|
| 3 | 0.1 |
| 7 | 0.3 |
| 11 | 0.5 |
| 15 | 0.7 |
| 21 | 0.9 |
| 27 | 1.2 |
| 101 | 4.6 |
| 201 | 9.1 |

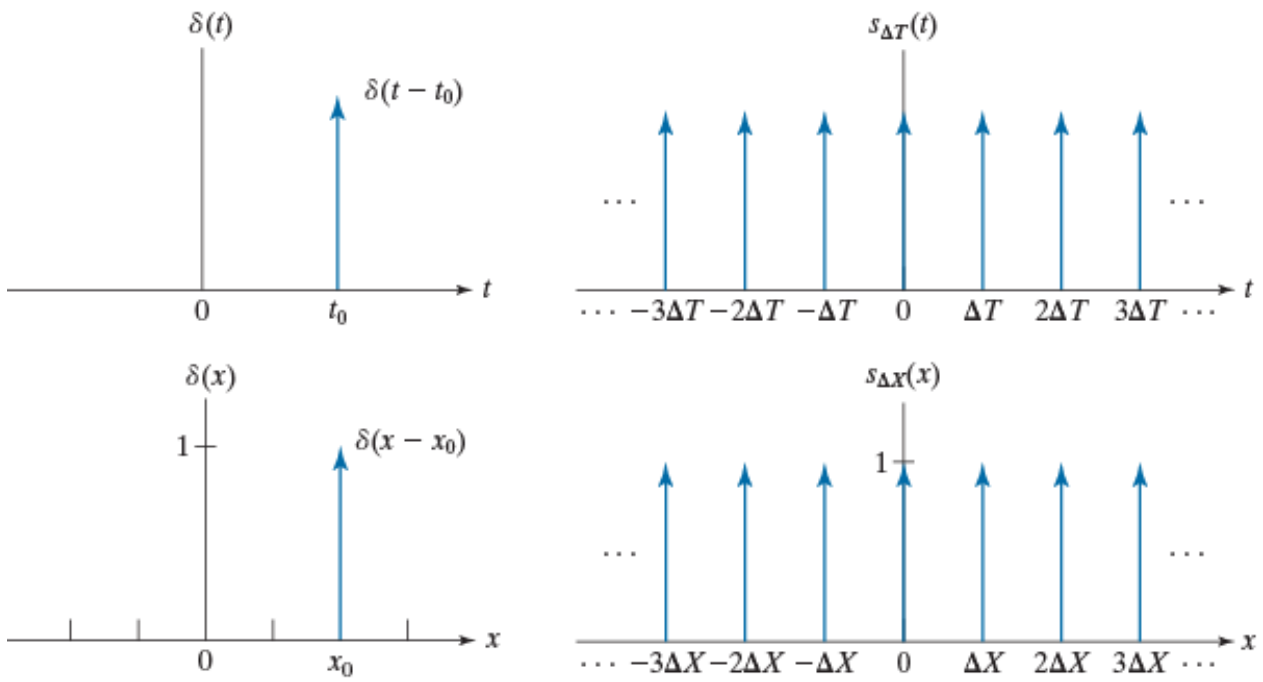
a b

**FIGURE 4.2** (a) Computational advantage of the FFT over non-separable spatial kernels. (b) Advantage over separable kernels. The numbers for $C(m)$ in the inset tables are not to be multiplied by the factors of 10 shown for the curves.

$$C_n(m) = \frac{M^2 m^2}{2M^2 \log_2 M^2} = \frac{m^2}{4 \log_2 M}$$

$$C_s(m) = \frac{2M^2 m}{2M^2 \log_2 M^2} = \frac{m}{2 \log_2 M}$$

# Preliminary Concepts (Unit Impulse)




a b
c d

**FIGURE 4.3** (a) Continuous impulse located at $t = t_0$. (b) An impulse train consisting of continuous impulses. (c) Unit discrete impulse located at $x = x_0$. (d) An impulse train consisting of discrete unit impulses.

# Preliminary Concepts (Box Function)

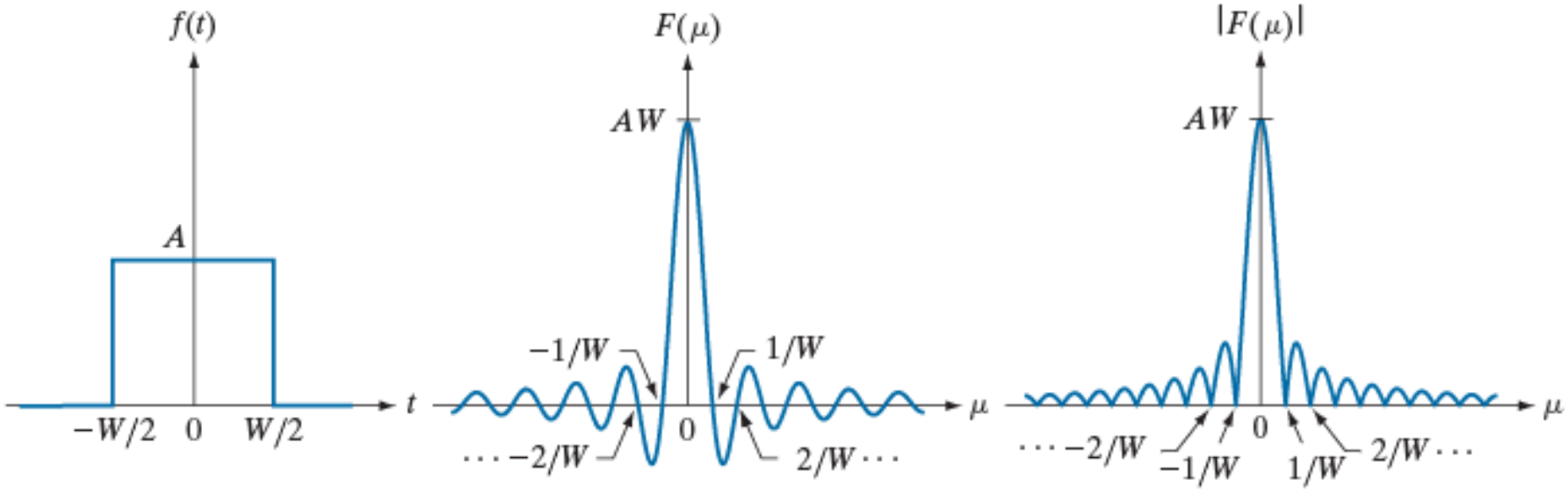
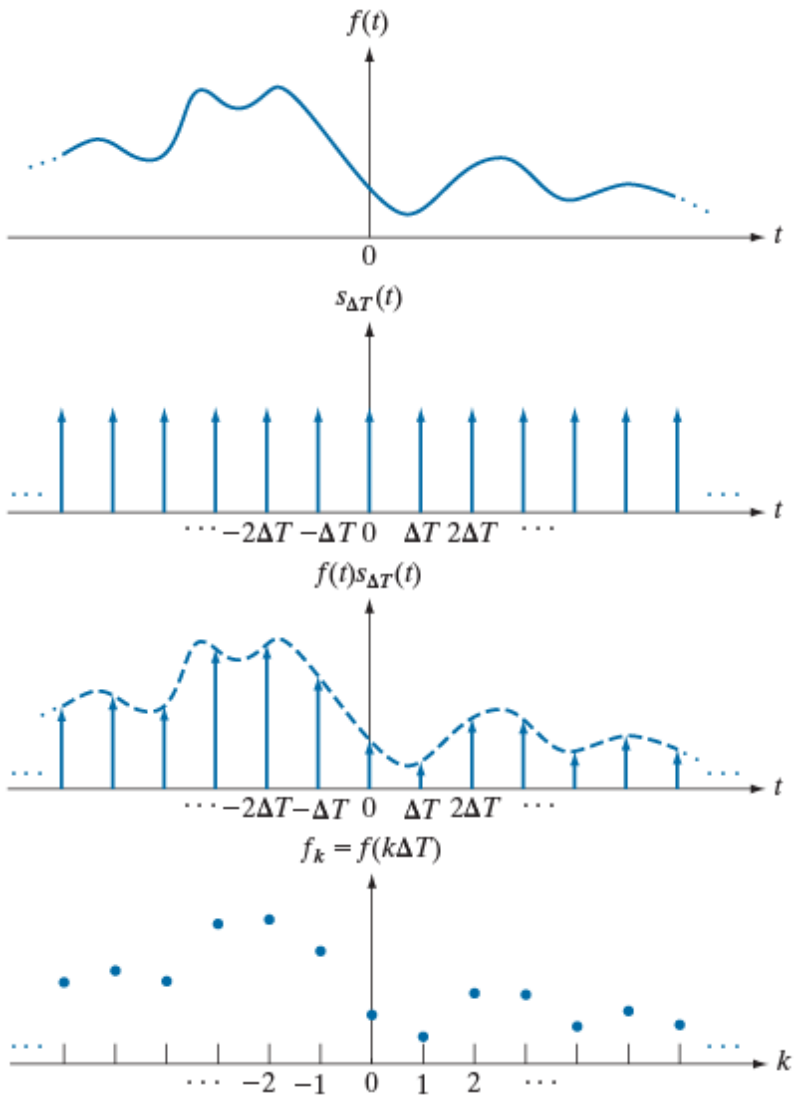



**FIGURE 4.4** (a) A box function, (b) its Fourier transform, and (c) its spectrum. All functions extend to infinity in both directions. Note the inverse relationship between the width, $W$, of the function and the zeros of the transform.

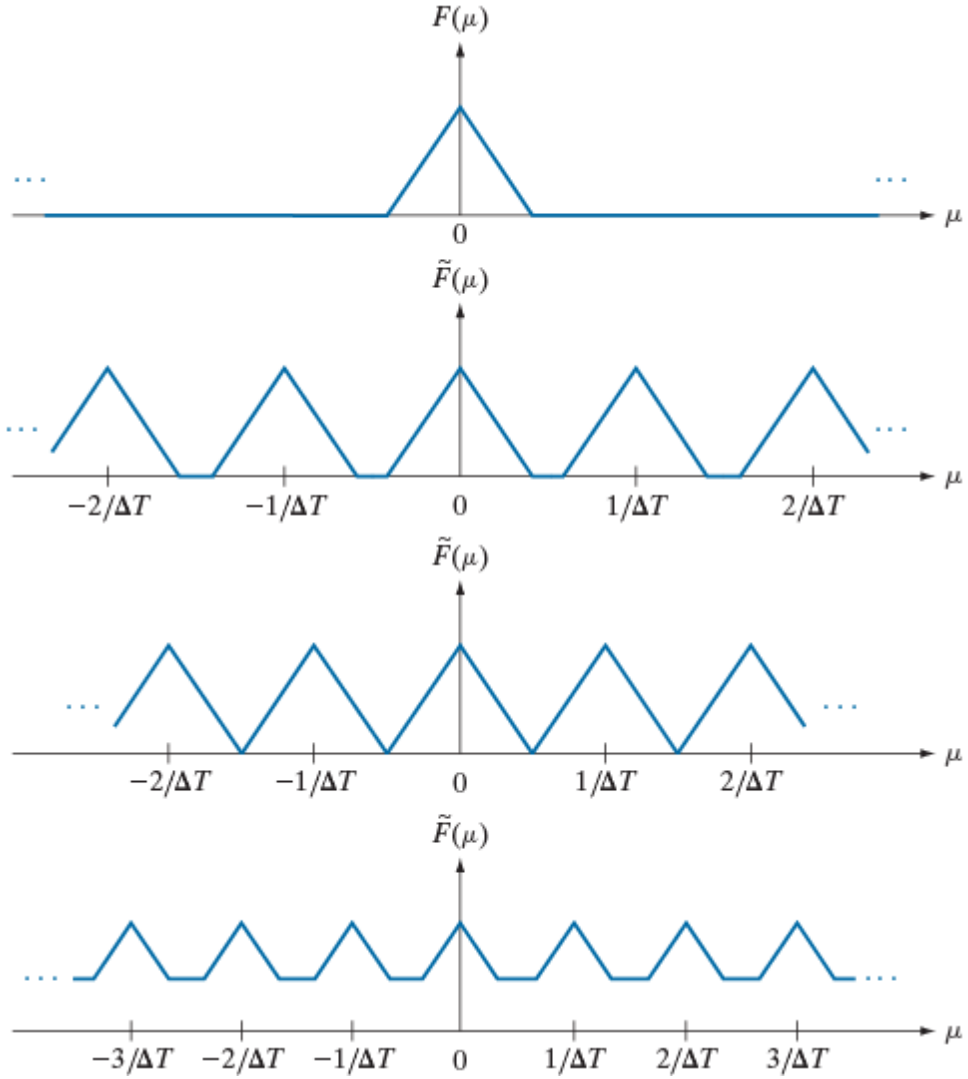
# Sampling




a
b
c
d

**FIGURE 4.5** (a) A continuous function. (b) Train of impulses used to model sampling. (c) Sampled function formed as the product of (a) and (b). (d) Sample values obtained by integration and using the sifting property of impulses. (The dashed line in (c) is shown for reference. It is not part of the data.)

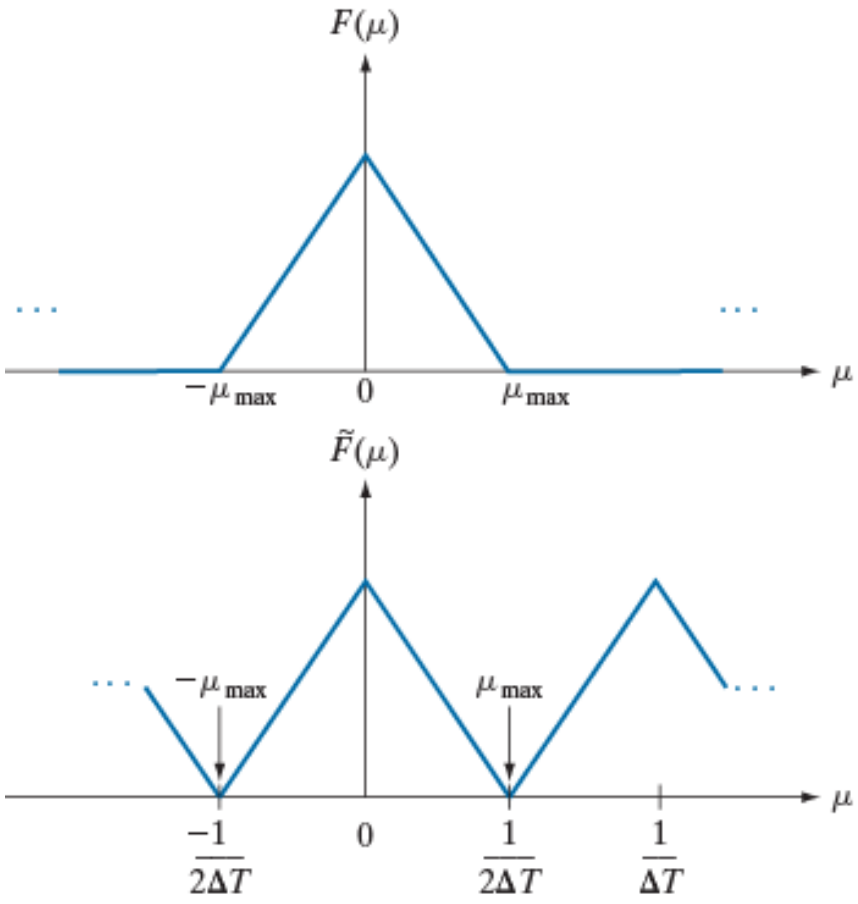
# Fourier Transform of a Sampled Function




a
b
c
d

**FIGURE 4.6** (a) Illustrative sketch of the Fourier transform of a band-limited function. (b)–(d) Transforms of the corresponding sampled functions under the conditions of over-sampling, critically sampling, and under-sampling, respectively.

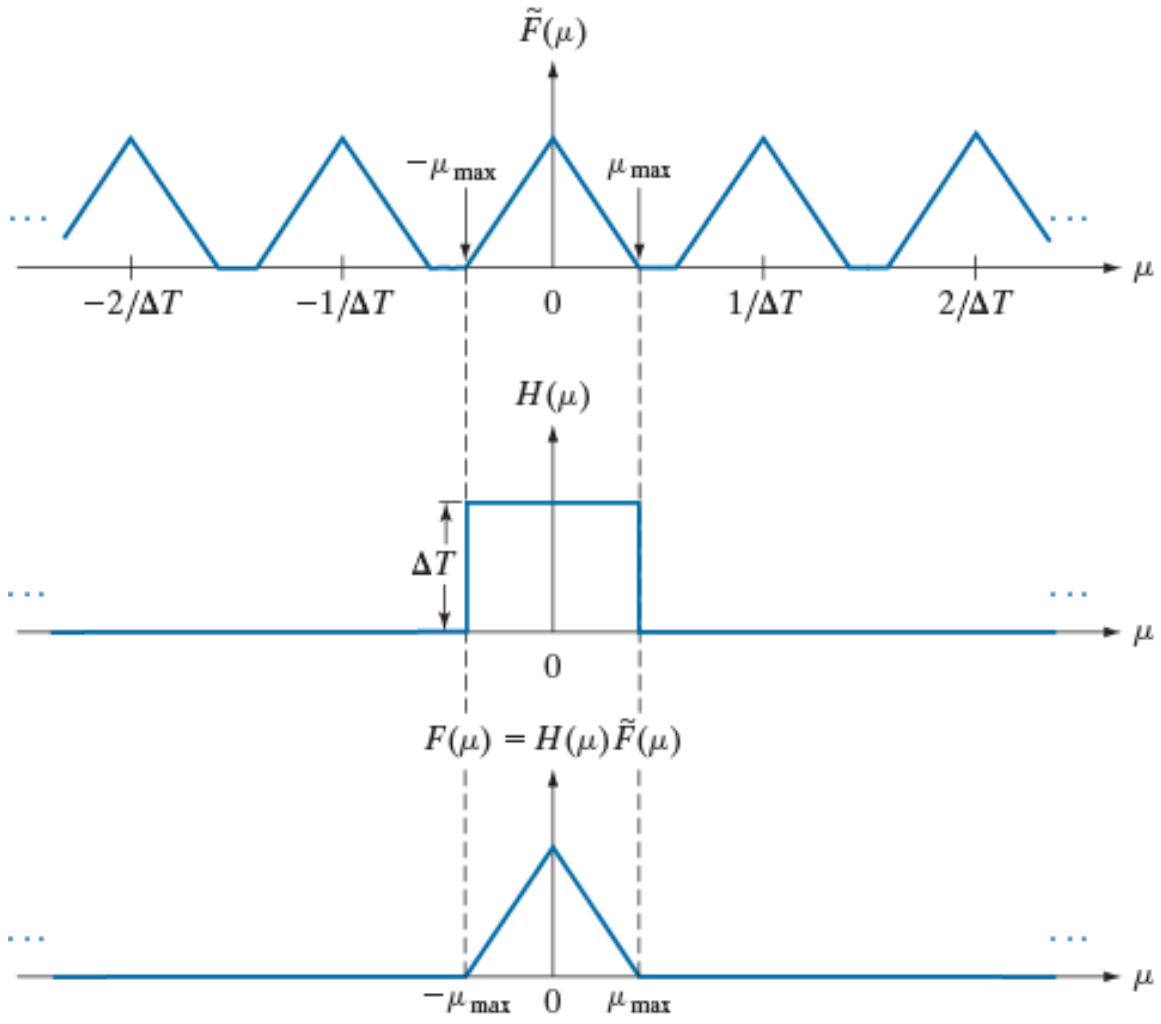
# Fourier Transform of a Sampled Function




a
b

**FIGURE 4.7** (a) Illustrative sketch of the Fourier transform of a band-limited function. (b) Transform resulting from critically sampling that band-limited function.

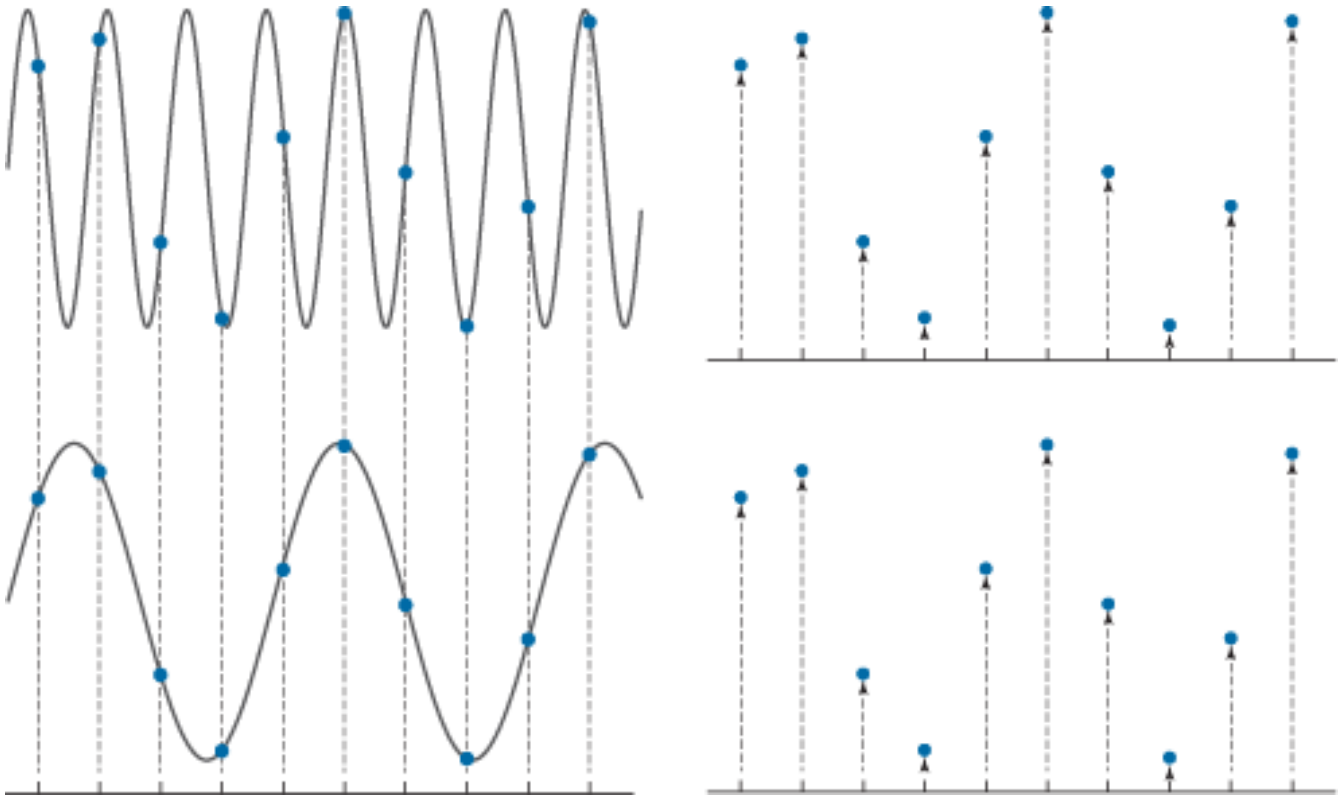
# Fourier Transform of a Sampled Function




a
b
c

**FIGURE 4.8** (a) Fourier transform of a sampled, band-limited function. (b) Ideal lowpass filter transfer function. (c) The product of (b) and (a), used to extract one period of the infinitely periodic sequence in (a).

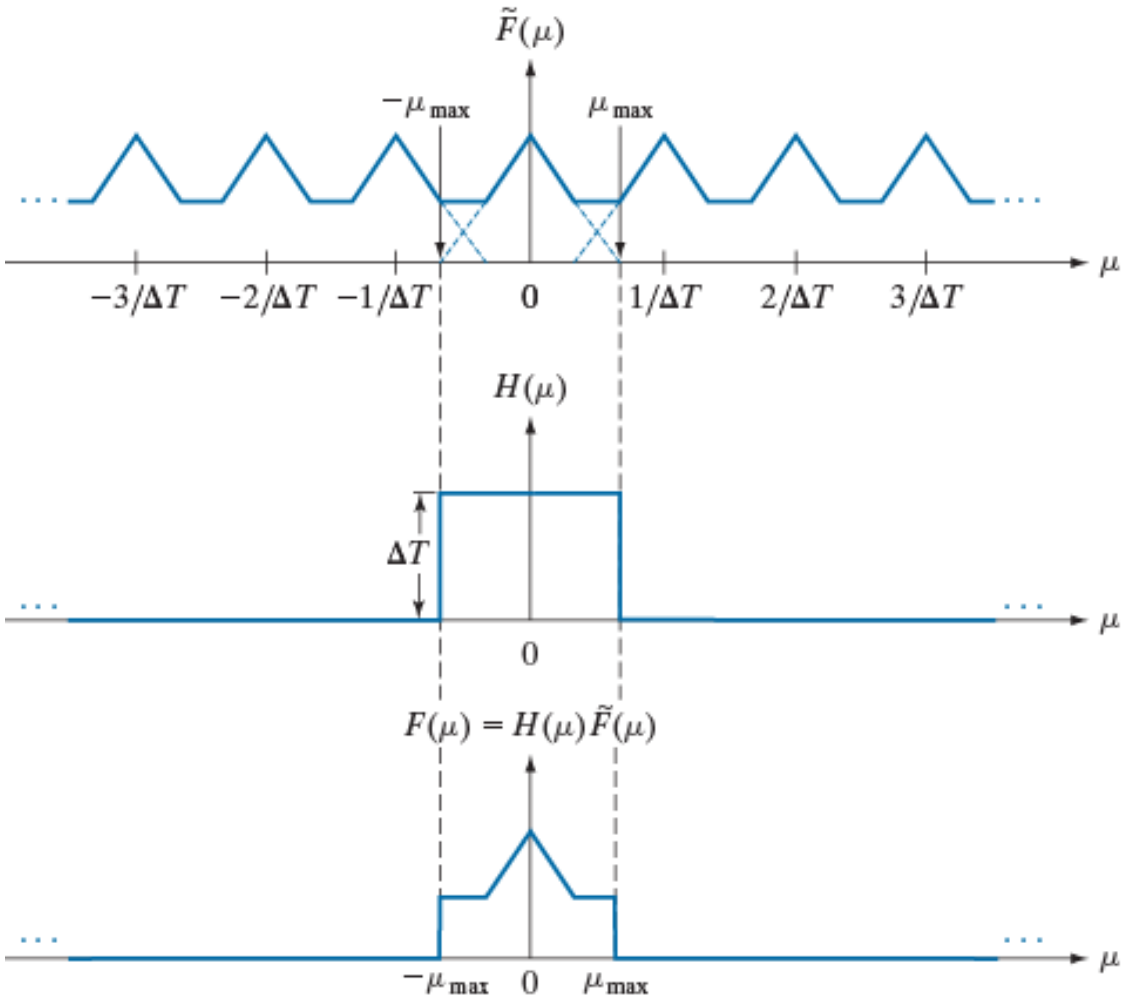
# Aliasing



a b
c d

**FIGURE 4.9** The functions in (a) and (c) are totally different, but their digitized versions in (b) and (d) are identical. Aliasing occurs when the samples of two or more functions coincide, but the functions are different elsewhere.

# Aliasing




a
b
c

**FIGURE 4.10** (a) Fourier transform of an under-sampled, band-limited function. (Interference between adjacent periods is shown dashed). (b) The same ideal lowpass filter used in Fig. 4.8. (c) The product of (a) and (b).The interference from adjacent periods results in aliasing that prevents perfect recovery of $F(\mu)$ and, consequently, of $f(t)$.

# Aliasing

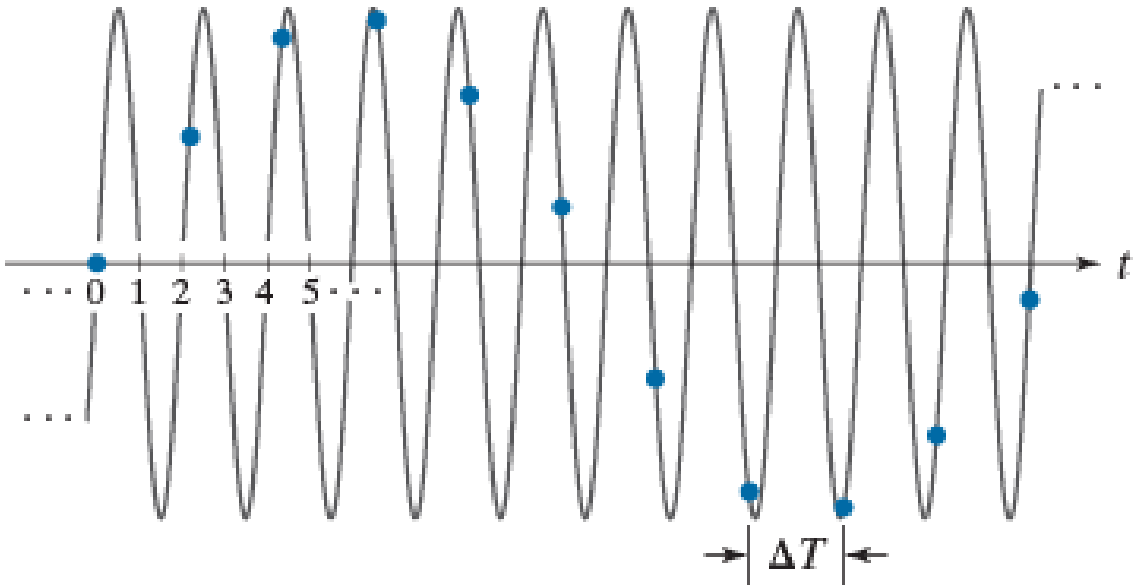
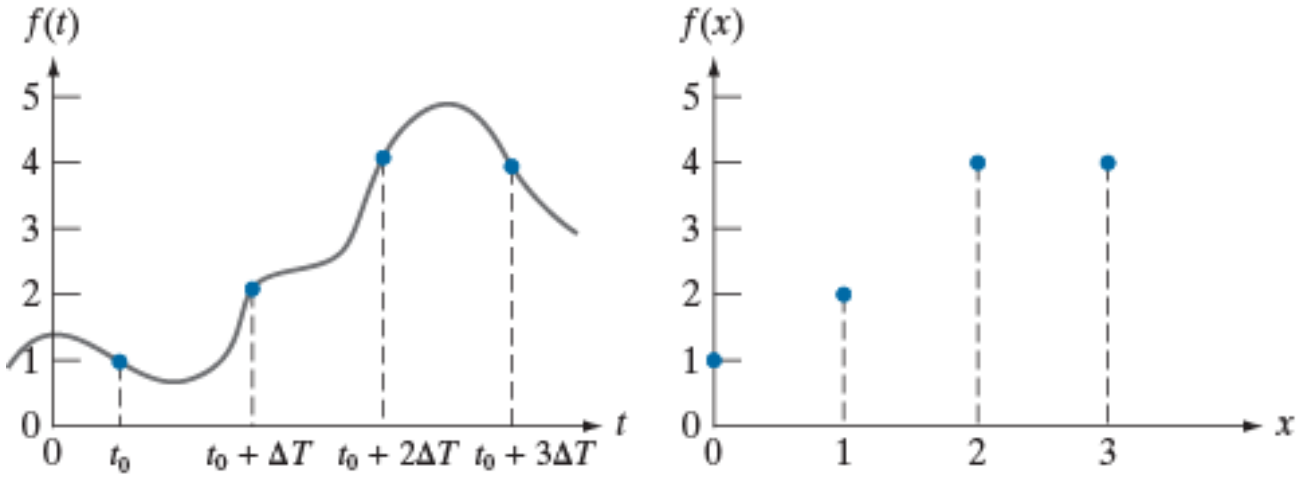



**FIGURE 4.11** Illustration of aliasing. The under-sampled function (dots) looks like a sine wave having a frequency much lower than the frequency of the continuous signal. The period of the sine wave is 2 s, so the zero crossings of the horizontal axis occur every second. $\Delta T$ is the separation between samples.

# Sampling Interval




a b

**FIGURE 4.12** (a) A continuous function sampled $\Delta T$ units apart. (b) Samples in the $x$-domain. Variable $t$ is continuous, while $x$ is discrete.

# 2-D Impulse

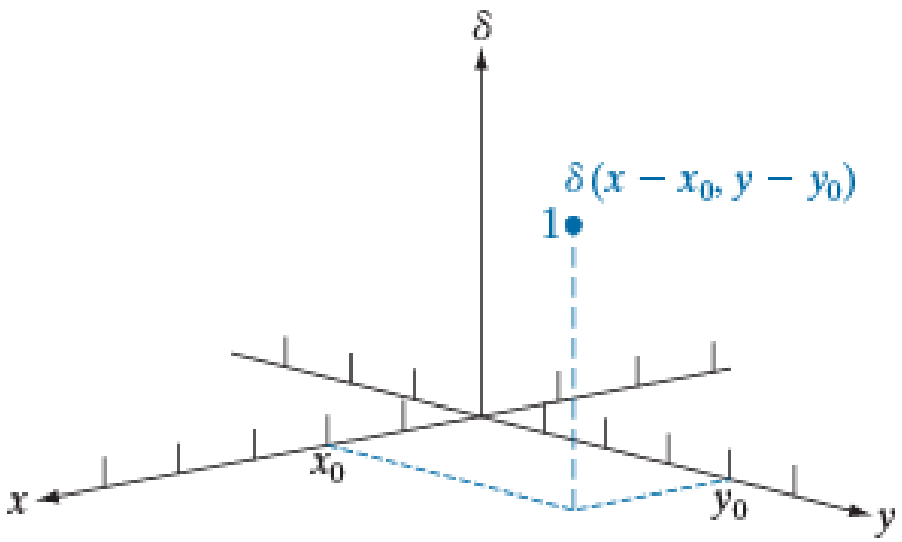
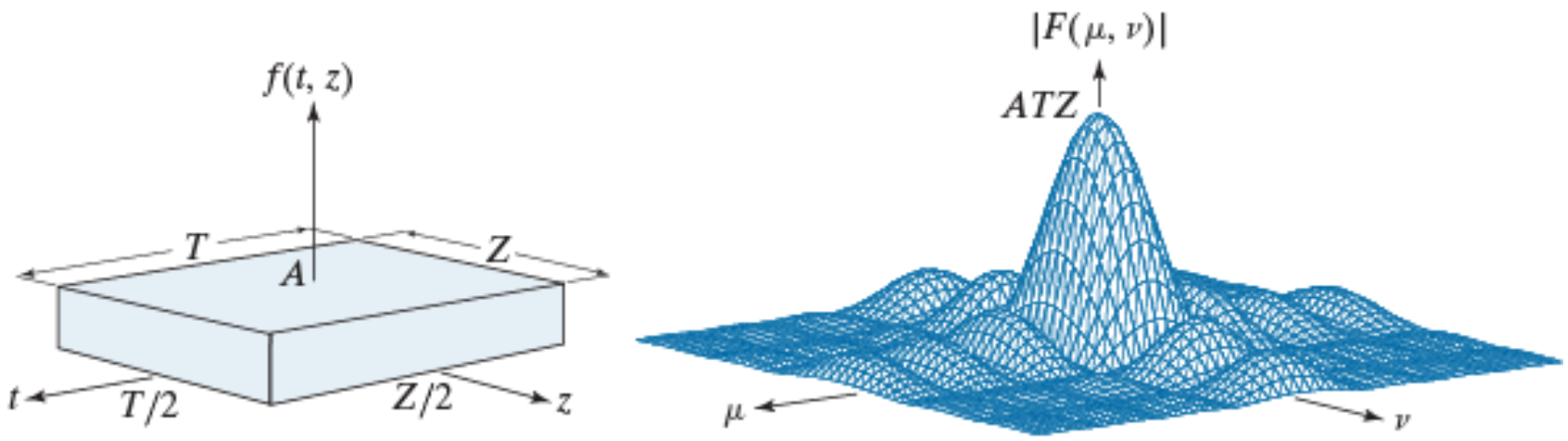



**FIGURE 4.13**
2-D unit discrete impulse. Variables $x$ and $y$ are discrete, and $\delta$ is zero everywhere except at coordinates $(x_0, y_0)$, where its value is 1.

# 2-D Box Function




a b

**FIGURE 4.14**
(a) A 2-D function and (b) a section of its spectrum. The box is longer along the $t$-axis, so the spectrum is more contracted along the $\mu$-axis.

# 2-D Sampling

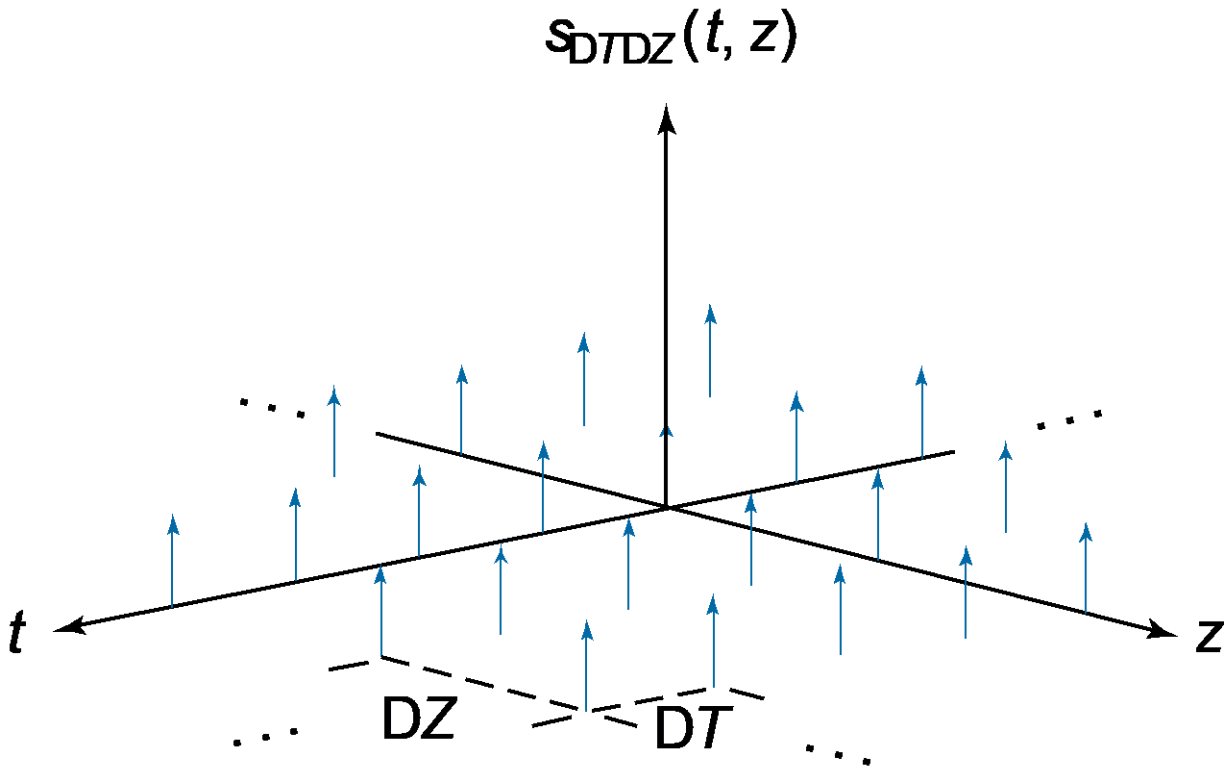
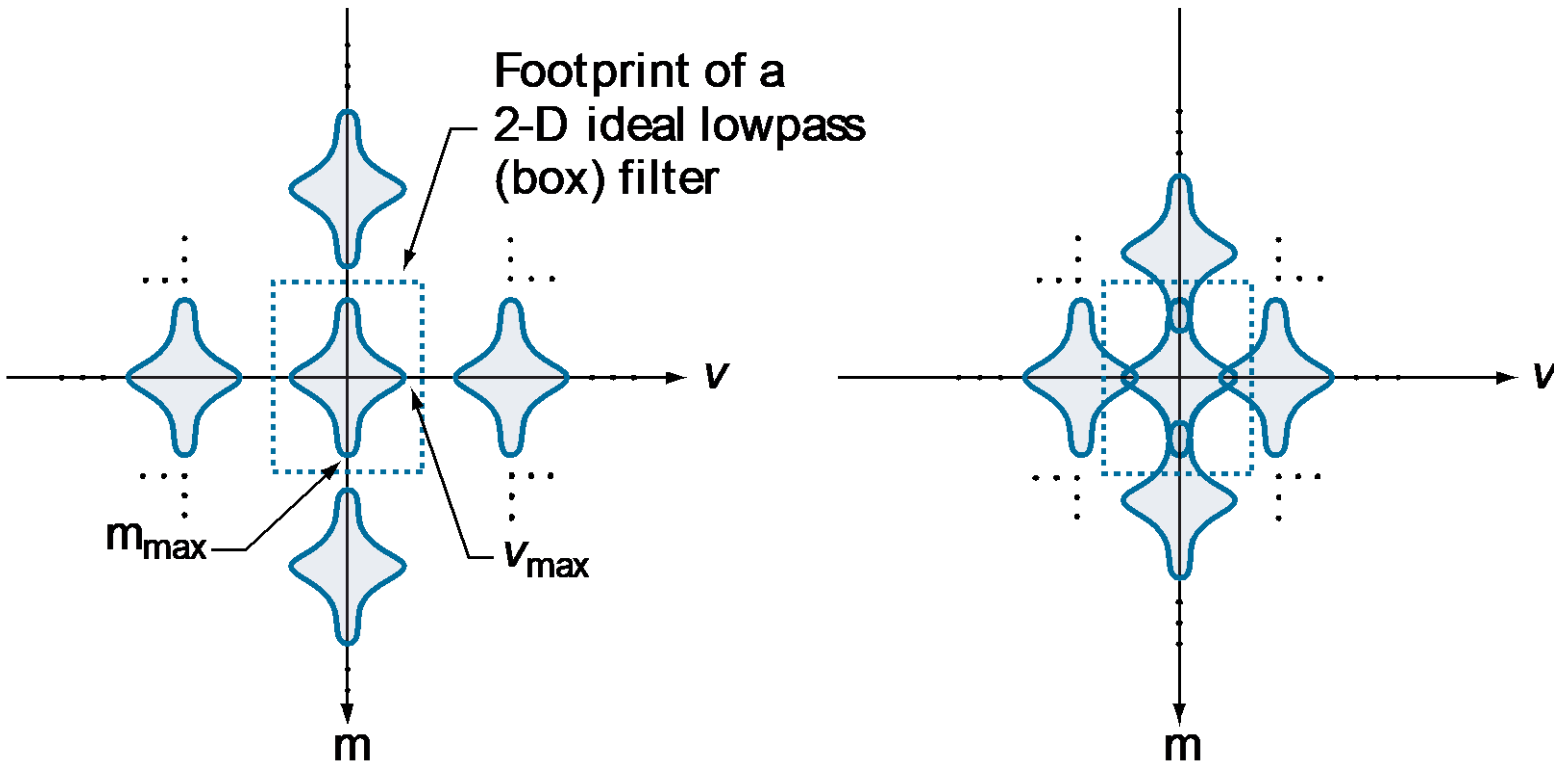



**FIGURE 4.15**
2-D impulse train.

# 2-D Sampling




a b

**FIGURE 4.16** Two-dimensional Fourier transforms of (a) an over-sampled, and (b) an under-sampled, band-limited function.

# Aliasing in Images

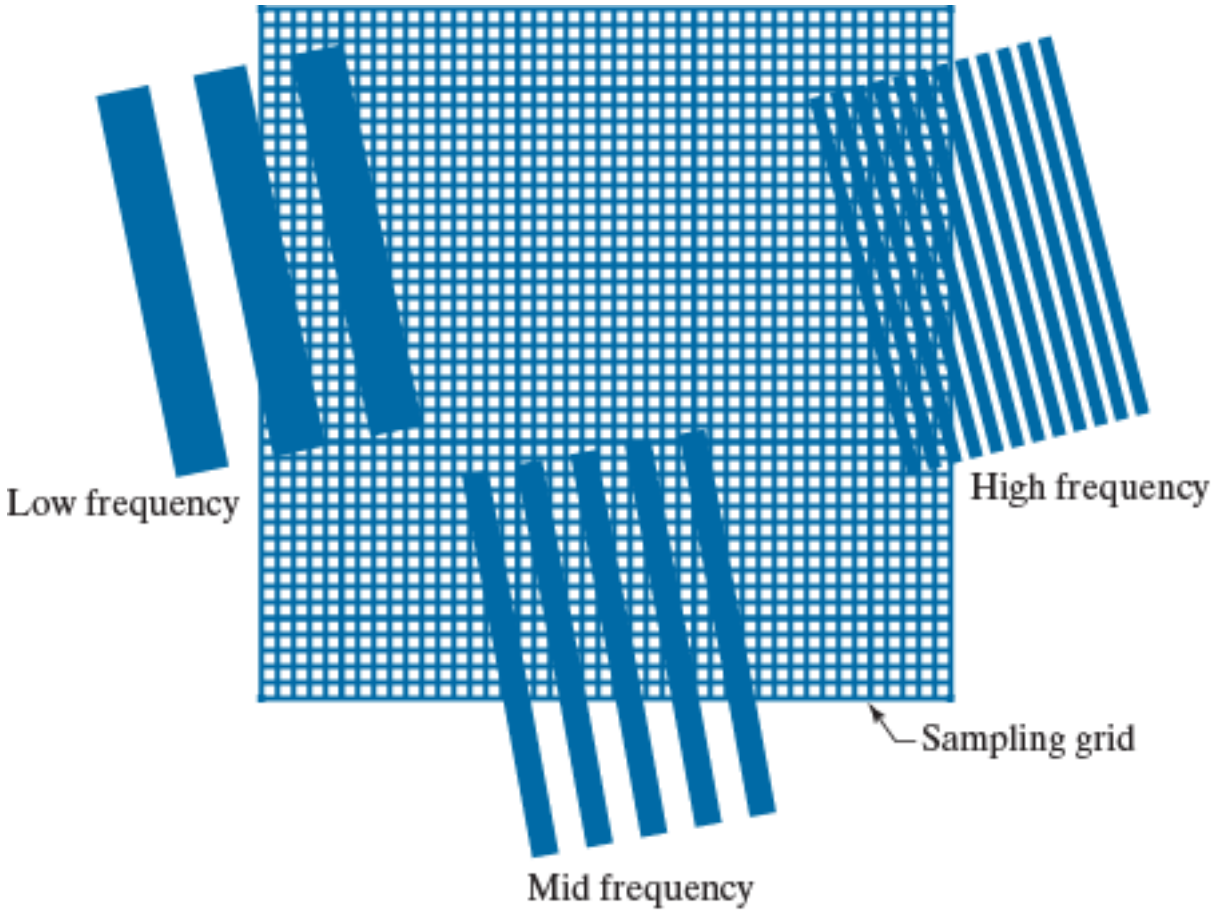
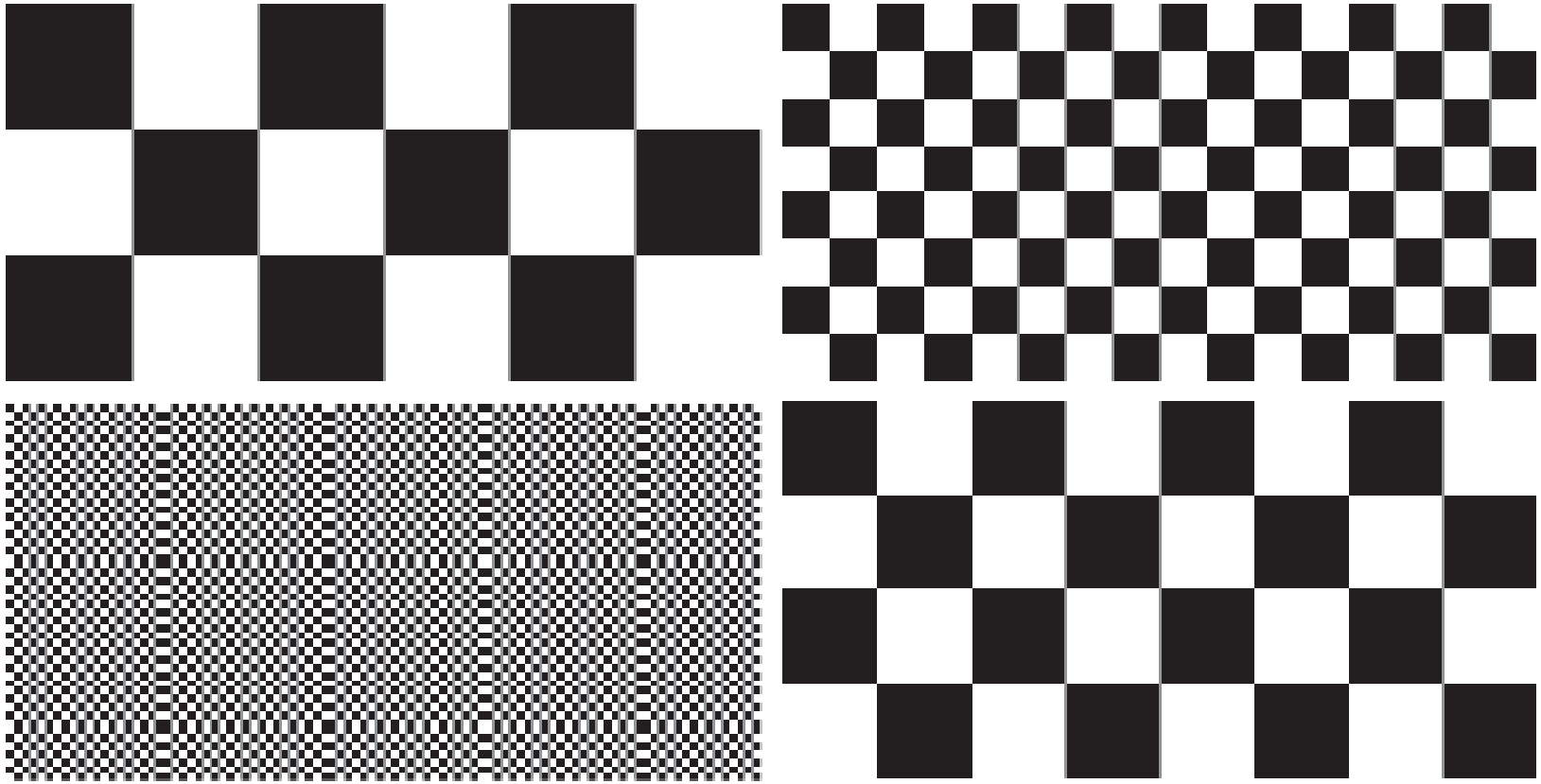



**FIGURE 4.17** Various aliasing effects resulting from the interaction between the frequency of 2-D signals and the sampling rate used to digitize them. The regions outside the sampling grid are continuous and free of aliasing.

# Aliasing in Images



a b
c d

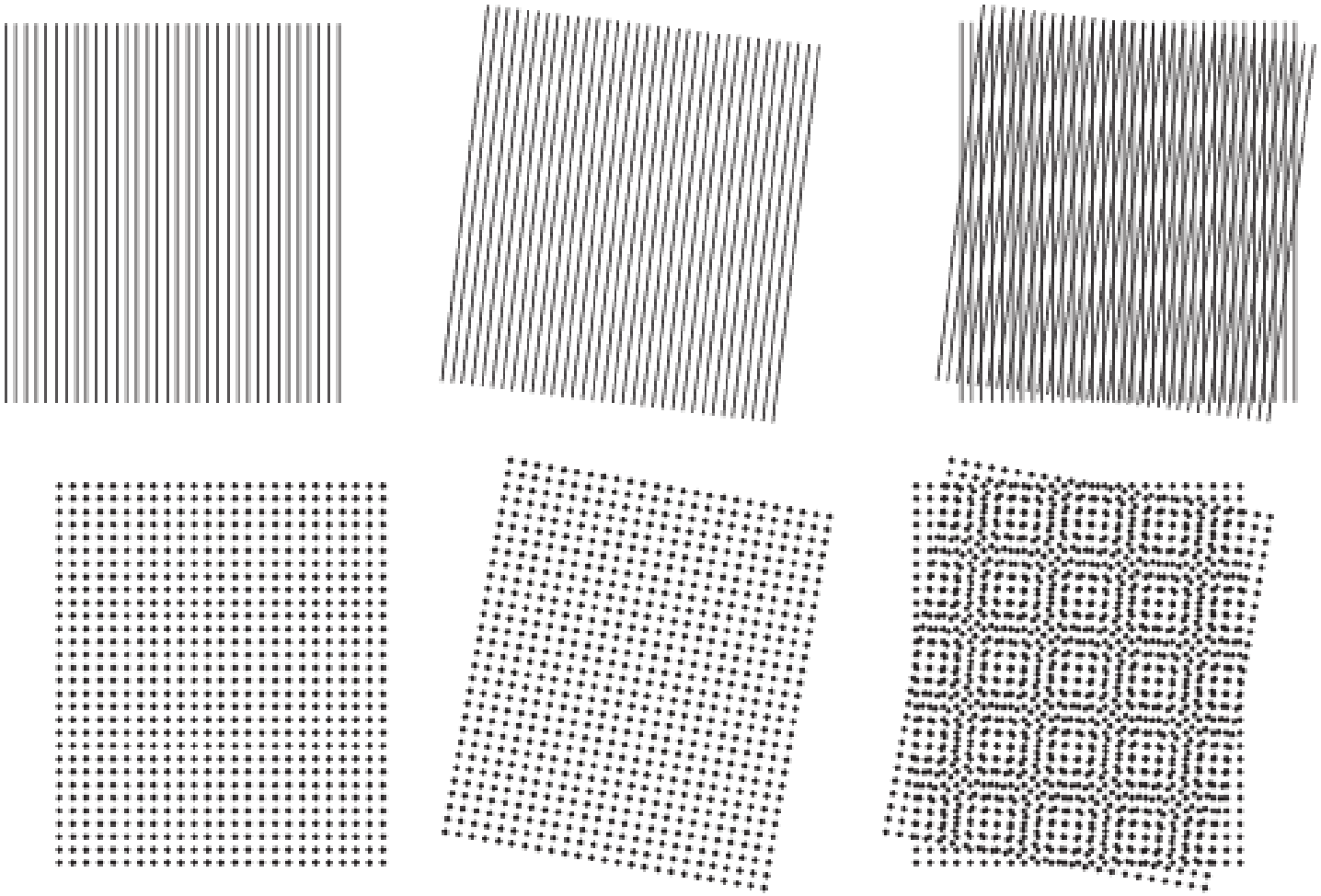
**FIGURE 4.18** Aliasing. In (a) and (b) the squares are of sizes 16 and 6 pixels on the side. In (c) and (d) the squares are of sizes 0.95 and 0.48 pixels, respectively. Each small square in (c) is one pixel. Both (c) and (d) are aliased. Note how (d) masquerades as a "normal" image.

# Aliasing in Images

a b c

**FIGURE 4.19** Illustration of aliasing on resampled natural images. (a) A digital image of size $772 \times 548$ pixels with visually negligible aliasing. (b) Result of resizing the image to 33% of its original size by pixel deletion and then restoring it to its original size by pixel replication. Aliasing is clearly visible. (c) Result of blurring the image in (a) with an averaging filter prior to resizing. The image is slightly more blurred than (b), but aliasing is not longer objectionable. (Original image courtesy of the Signal Compression Laboratory, University of California, Santa Barbara.)

# Moire Patterns



a b c
d e f

**FIGURE 4.20** Examples of the moiré effect. These are vector drawings, not digitized patterns. Superimposing one pattern on the other is analogous to multiplying the patterns.

# Moire Patterns

**FIGURE 4.21** A newspaper image digitized at 75 dpi. Note the moiré-like pattern resulting from the interaction between the ±45° orientation of the half-tone dots and the north-south orientation of the sampling elements used to digitized the image.